CURRENT FILE

Bt102

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1. How we can write scientific name of organisms 2

Linnaeus (in 1735) established the system of scientific nomenclature. Each organism has two names: the genus and specific epithet (descriptive word).
Are italicized or underlined. The genus is capitalized, and the specific epithet is lowercase. Escherichia coli

2. write name of five types of microscopy 8

- Darkfield microscopy
- Phase contrast microscopy
- Fluorescent microscopy
- Transmission electron microscopy
- Scanning electron microscopy

3. Enlist step of gram staining 10

- Applying primary stain (crystal violet).
- Adding mordant (Gram's iodine).
- De-colorization with ethanol, acetone or a mixture of both.
- Counterstaining with safranin.

4. Acid fast procedure 11

Some bacteria such as Mycobacteria have a waxy material in their cell wall. Specific name for this waxy substance is mycolic acid

Principle of Acid-Fast Staining:

Mycobacteria are lipophilic, not easy to stain but once stained, they are resistant to acidic alcohol decolorization process.

Procedure for Acid-Fast Staining:

- Make a smear
- Carbolfuchsin on slides
- Heat the slide for 5-10 min
- Wash with water and then with acid alcohol for 1 min.
- Counterstain with methylene blue for 1 min
- See under the microscope.
- Acid fast bacteria have mycolic acid that can be stained by heating the organisms with the stain

5. Difference between prokaryotes and eukaryotes 12

PROKARYOTES	EUKARYOTES
Prokaryotes are unicellular	Eukaryotes are multi-celled organisms
DNA is stored in the cytoplasm of prokaryotes.	DNA is stored within the nucleus of eukaryotes
Nucleus is absent	Nucleus is present

Membrane bound organelles is absent	Membrane organelles is present
One chromosome is present, but not true	More than one number chromosomes present
Examples: bacteria and archaea	Examples: protist, plants, fungi, animal

6. Shapes of bacteria 13

Three types:

- **Coccus:** These are circular or spherical in shape.
- **Bacillus:** These rod shaped bacteria.
- **Spiral:** These are curved shaped bacteria. They are further divided into 3 more subgroups.
  - **Vibrio:** curved rods
  - **Spirillum:** Helical but rigid
  - **Spirochete:** Helical but flexible Move by axial filaments

7. Fimbriae location & function 16

- Hair-like small appendages on G-neg cells
- Composed of pilin, a protein
- Can be at one pole or around the entire cell
- Used for attachment to surfaces of epithelial cell
- Fimbriae help bacteria attach to surfaces
- Fimbriae also help attach to epithelial cells e.g. Neisseria gonorrhoeae

8. Three components of lipopolysaccharides? 18

- Lipid A: endotoxin – Fever, vasodilation and shock
- Core polysaccharide: Structural support
- O side chain: Antigenic like teichoic acid in gram positive bacteria

9. Bacterial classes in on the basis of temperature 27

- **Psychrophiles:** Cold loving 15°C
- **Mesophiles:** Moderate temp loving 25 – 40°C, Optimum 37°C
- **Thermophiles:** Heat loving 50 – 60°C

10. Enlist four phases of bacterial growth curve 33

- **The Lag Phase:** initial period of bacterial cells in new environment before starting growth.
- **The Log Phase:** in this phase bacteria growth occur and cell number increase
- **The Stationary Phase:** In the stationary phase, growth reaches a point in which number of dying cells equals the number of dividing cells. This is also called a period of equilibrium
- **The Death Phase:** The number of death cells increase and number of new cells decreased in this phase is called death phase.

11. Which factors can influence the bacterial death rate 33

The reason is simple: Nutrients are depleted and waste products which are toxic to the cells accumulate suppressing the growth and killing the cells.

The rate of growth or death of a particular microbial species is influenced by a variety of physical factors in its environment including temperature, osmotic pressure, pH, and oxygen concentration.

12. Sepsis, antisepsis , sanitization 35

- **Sepsis:** Sepsis is the extreme response of body to an infection
- **Antisepsis:** Removing pathogens from living tissue
- **Sanitization:** removal bacteria from surface

13.Plasma membrane disrupt 40

Hexacholorphene, triclosan, Chlorhexidine, Alexidine

14.Dry heat method? 42

- Direct flaming
- Incineration
- Hot air sterilization(hot-air oven)

15.Define, Gene, Genome, Chromosomes and phenotypes 44

Gene: The basic unit of heredity passed from parent to child
Genome: The genome is the entire set of DNA found in a cell.
Chromosome: threadlike structures made up of protein and DNA that carry genomic information from cell to cell.
Phenotype: all observable characters that influenced by genotypes and environment

16.Classification of five kingdom system 48

In 1969 the five kingdom classification was proposed by Robert whittakar

- **Monera**: bacteria and archaea
- **Plantae:** plants
- **Animalia :** Animals
- **Protista:** they are unicellular eukaryotes.
- **Fungi:** yeast mold mushrooms, consist of chitin

17.Detail note on western blotting method 53

We can use this technique for the detection of antigen in the serum. Microbial protems can be separated on SDS-PAGE by electrophoresis and the presence of these proteins can be detected by enzyme-tagged antibodies specific to those proteins. A color band will be seen where the specific protein (antigen) 1s present on the gel. Please remember that proteins in the gel are first transferred to a paper strip before they could be detected by specific antibodies

18.Sexual reproduction in fungi 58

Sexual Spores

- **Plasmogamy:** A haploid nucleus of a donor cell (+) penetrates the cytoplasm of a recipient cell (-).
- **Karyogamy:** The (+) and (-) nuclei fuse to form a diploid zygote nucleus.
- **Meiosis:** The diploid nucleus gives rise to haploid nuclei (sexual spores)

19.Basic features of yeast 59

- Non-filamentous fungi
- Unicellular Reproduce by budding (asexually) or sexually
- Some are dimorphic
- Larger than bacteria
- No flagella or cilia
- Facultative anaerobes
- Gram positive

20.Importance of fungi 61

Saccharomyces are used for alcoholic beverages under anaerobic conditions for alcohol production (fermentatively). However, if Saccharomyces are incubated aerobically (for bread making), they metabolize glucose to produce CO2 (some ethanol is also produced but gets evaporated during baking) which expands the dough causing it to rise. This yeast is also extensively used for molecular biology work and vaccine production. Vitamin C is obtained from Aspergillus niger; cellulase from Trichoderma, antibiotics from many fungi and anticancer Taxol from Taxomyces are some other examples of useful fungal products. Entomophaga (kills gypsy moth), hence is used as a biological pest control. Mushrooms can be cooked and eaten as a source of proteins for humans.

21.Consequences of viral infections 68

- Lysis of cells
- Microscopic and macroscopic Degenerative changes (Cytopathic Effects)
- **Virulent infection:** lysis of cells
- **Latent infection:** Virus does not replicate
- **Persistent infection:** Virus replicates but slowly
- **Transformation:** Tumor cells

22.Microbiota Factor And Distribution Composition 72

Availability of various nutrients determines the type of organisms present in a system. Similarly, physical and chemical factors also influence the type of microbiota. Other factors include pH, O2, and CO2.

23.Types of Transmission of disease 74

- **Direct Transmission:** Person to person physical contact is a direct contact. Influenza spread is a typical example.
- **Indirect Transmission:** A nonliving object is involved. Objects in the use of a patient such as handkerchief, utensils, pillow and bedding are called fomites. These can transmit organisms to susceptible individuals.
- **Droplet Transmission:** Mucus droplets are created when you sneeze for example. These droplets carry organisms that may infect another individual who comes in contact with the droplets. One sneeze may produce 20,000 droplets. Influenza can be spread by this route.
- **Vehicle Transmission:** Transmission of disease agents by a medium, such as water, food, or air. Other media include blood and other body fluids, drugs, and intravenous fluids
- **Mechanical Transmission:** In this type of transmission, organisms stick to the body parts of insects and get transmitted passively from one location to another.
- **Biological Transmission:** Microbes multiply in the insect and is transmitted actively by insect bites from one individual to another

24.Explain natural and acquired immunity and active, passive immunity 78

Ability of the body to protect itself from harmful effects of diseases

Two types:

- **Innate Immunity (natural):**

Plants and animals have what is called mnate immunity. Innate immunity is the first line of defense against pathogens. It involves several cell types, proteins, and even an organ. The organ

involved is your skin. Yes, skin is part of the first line of defense. It protects you and prevents pathogens from getting inside your body.

Naturally Acquired Active Immunity: This type of adaptive immunity develops in a person who gets infected with a microbe but then recovers from that illness making antibodies.

Naturally Acquired Passive Immunity: A typical example of this type of immunity is transfer of antibodies from colostrum or milk of a mother to the offspring.

- **Adaptive immunity (acquired):**

Adaptive immunity is an important part of the immune system. It is protection from an infectious disease agent that is mediated by B- and T- lymphocytes following exposure to specific antigen, and characterized by immunological memory

Artificially Acquired Active Immunity: When a person receives a vaccine against a microbe and the body makes antibodies in response to that vaccine, it is called artificial acquired active immunity because body makes these antibodies actively by itself.

Artificially Acquired Passive Immunity: When premade antibodies are given to a person in order to protect a person from an illness or a snake bite, this constitutes an artificially acquired passive immunity type. The following figure illustrates these four types

25.What is PAMPS? 79

Pathogen associated molecular patterns (PAMPs): Formed by LPS, flagellin, peptidoglycan, DNA, RNA

26.Inflammation and its consequence? 80

Damage to body tissues triggers a local defensive response which is called inflammation. It causes:

- Redness
- Pain
- Swelling

Heat During inflammation, various mediators of inflammation is released, some of which include:

- Histamine
- Kinins
- Prostaglandins by damaged cells
- Leukotrienes by mast cells
- Cytokines are released (TNF-alpha) o Cause’s release of acute phase proteins from liver.

27.Interferon and it's types 80

These are antiviral proteins, produced by macrophages and lymphocytes in response to viral infections. Fibroblasts can also secrete it. These proteins interfere with viral replication.

Species specific: This means that interferon produced by humans cannot be used in dogs or cats or vice versa.

Not viral specific: This means that interferon produced in response to virus A can also be effective against virus B.

Its types:

- **Alpha:** It is released by all infected cells.
- **Beta:** Also produced by infected cells.
- **Gamma:** This one is released by lymphocytes only. Lymphocytes belong to adaptive immune system (just remember that).

28.Antigen and epitope 82

Antigen: A foreign substance that provokes immunity is called an antigen. The definition of antigen is a harmful substance which enters the body which causes the body to make antibodies as a response to fight off disease. More specifically called as an immunogen. Proteins are best antigen, though glycoproteins and lipoproteins can also act as antigens. Carbohydrates and lipid alone are poor antigens.

Epitopes: Specific regions on an antigen. In other words, any antigen has multiple regions each one of those acts as a small antigen. These are called epitopes. These epitopes are also called as antigenic determinants. Most antigens have MW >10,000 D. Molecules that are less than 10,000 D are generally poor antigen.

29.Primary & secondary immune response 86

When a person is exposed to an antigen (a microbe for example), the antigen is presented to B cells which start making antibodies. Initially, there are no detectable antibodies in the serum. However, after 3-5 days, IgM antibodies are secreted and then 7-10 days after, the same B cell starts making IgG. The level of IgG rises to a level and then it starts declining. This is a typical response of an antigen exposure and is called a primary immune response.

It is interesting to note that a second exposure with the same antigen results in a heightened response in terms of very high levels of antibodies that are achieved in a very short time (within 2-7 days). This is called a secondary immune response. It is also called a memory or anamnestic response because memory cells are involved that ensure a quick response to the second exposure by the same antigen

30.Hypersensitivity and types 92

It is a normal Immune response but in a damaging way. In other words, immune system is acting in a way for which it has made to. So, when an antigen gets into the body, immune system recognizes it as foreign and start making antibodies against the antigen. Allergy is another name for hypersensitivity. Those antigens that act in a bad way are called allergens

Types:

- **Type I (Anaphylactic):** Also called anaphylaxis or immediate hypersensitivity. This is IgE mediated. Happens in<30 mints
- **Type II (Cytotoxic):** IgG and IgM mediated as these antibodies attach to cells and destroy them by activating complement
- **Type III (Immune complex mediated):** Immune complexes are formed by IgM or IgG antibodies and these complexes lodge in the capillary beds and cause damage by complement activation.
- **Type IV (delayed type cell-mediated):** CD4+ helper cells along with macrophages are involved.

31.Anaphylaxis ,causes, diagnosis & treatment 92

Anaphylaxis is a severe, potentially life-threatening allergic reaction. It can occur within seconds or minutes of exposure to something you're allergic to, such as peanuts or bee stings

- **Causes:** The patient also experiences difficulty in breathing. It is an emergency as it may kill the person. Anaphylaxis may be local or systemic. Shock from drug reactions, venoms, and common allergens causing asthma are some examples of type I hypersensitivity.

- **Diagnosed:** Diagnosed by demonstration of IgE by intradermal injections of allergens one by one. If specific IgE are present in the serum, a reaction will be visible on the skin locally in the form of a wheel of inflammation.
- **Treatment:** Epinephrine & antihistaminic drugs provide a means of treatment.

32.Antibiotics 97

Antibiotics are produced by microbes, but only a few are selectively toxic to microbes. Antibiotics are products of microbes against microbes. Must not be toxic to the body

33.Inhibits Growth Cell Wall Drug Name? 98

Penicillin's and cephalosporins are the major antibiotics that inhibit bacterial cell wall synthesis.

34.Disc diffusion method

The method consists of placing paper disks saturated with antimicrobial agents on a lawn of bacteria seeded on the surface of an agar medium, incubating the plate overnight, and measuring the presence or absence of a zone of inhibition around the disks

35.Microbes enter the body through which source

Microbes enter the body through ingestion, inhalation, direct contact, cutaneous penetration, and ascending infection

1. Steps of preparing stain of western bloating?